INPUT:

Network packets transmitted through the network.

OUTPUT:

Packet sniffer started

Source IP : 192.168.1.10
Destination IP : 142.250.183.14.
Protocol : 6

Source IP : 192.168.1.10
Destination IP : 142.250.72.14
Protocol : 17.

Exp: 14

PACKET SNIFFING AND RAW SOCKETS.

AIM:

To write a program using RAW sockets to implement packet sniffing.

ALGORITHM:

1. Create a raw socket.
2. Bind the socket to the network interface.
3. Receive network packets using the RAW socket.
4. Extract the packet header information.
5. Display the source IP, destination IP and protocol.
6. Continue capturing packets and display the details.

CODE:

```
import socket
import struct
s= socket.socket(socket.AF_INET, socket.SOCK_RAW,
                 socket.IPPROTO_IP)
print("Packet sniffer started....")
While True:
    packet, addr = s.recvfrom(65565)
    ip_header = packet[0:20]
    iph = struct.unpack('!BBHHHBBH4s4s', ip_header)
    source_ip = socket.inet_ntoa(iph[8])
    destination_ip = socket.inet_ntoa(iph[9])
    protocol = iph[6]
    print("Source IP:", source_ip)
    print("Destination IP:", destination_ip)
    print("Protocol:", protocol)
    print("-" * 40)
```

RESULT:

Thus, the program to implement packet sniffing using RAW sockets was executed successfully.